

Docker Assignment



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Assignment

- Take an existing Dockerfile and optimize it for build speed and image size.
- Use multi-stage builds where appropriate.
- Set up a custom Docker network and connect multiple containers to it.
- Use Docker commands to inspect and manage the network.
- Create and manage Docker volumes to persist data for your containers.
- Use named volumes and bind mounts and demonstrate how they can be used in a multi-container setup.
- Backup and restore data from Docker volumes.
- Implement security best practices in your Docker setup, including user permissions, image vulnerability scanning, and secret management.
- Configure Docker to run containers with the least privilege.
- Use tools like Docker Bench for Security to audit your Docker environment.

Requirements

- Docker 20.10 or later
- Docker Compose 1.29 or later
- Advanced Linux command-line knowledge
- GitHub account

Note

- Complete the tasks as specified.
- Submit your project in a GitHub repository.
- Ensure your submission is well-documented.
- Include a README file with instructions on how to set up and run your project.

Deliverables

- GitHub repository link with all source code and documentation.
- README file with setup instructions.
- Well-documented code with comments.
- Dockerfiles, Docker Compose files.

Bonus Question

- Implement a log aggregation system for your Dockerized application to collect, parse, and visualize logs, providing insights into the performance and behaviour of your application.